



Croatian Open Competition in Informatics

Round 6, March 21st 2026

Tasks

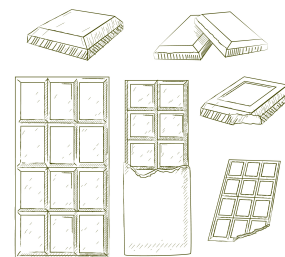
Task	Time limit	Memory limit	Score
Čokolada	1 second	512 MiB	50
Džeparac	1 second	512 MiB	70
Prepisivanje	1 second	512 MiB	110
Skijanje	1 second	512 MiB	110
Učionica	1.5 second	512 MiB	110
Total			450



Task Čokolada

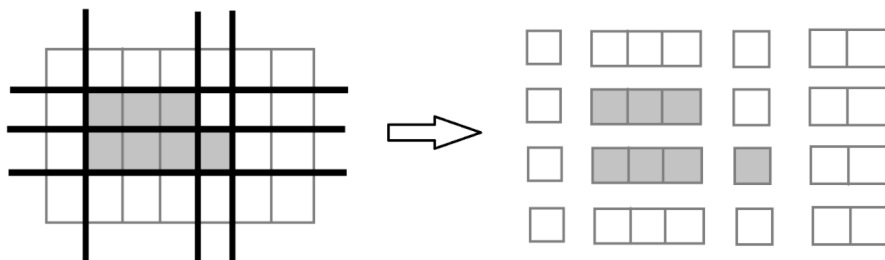
Luka loves chocolate. He is very excited to eat a large chocolate bar made of n rows and m columns he got for his birthday. The chocolate consists of black and white squares of chocolate. However, Luka doesn't really like white chocolate and would only like to eat the black squares.

Before he starts eating, Luka will cut the chocolate. He will make a number of vertical and/or horizontal cuts in between the rows and the columns of the chocolate. A vertical cut goes from the top edge of the chocolate to the bottom edge, while a horizontal cut goes from the left edge to the right edge. After making the cuts, Luka will obtain several rectangular pieces of chocolate.



Since Luka will only eat the black parts of the chocolate, he wants to completely separate the black squares from the white squares. This means that every resulting piece must consist entirely of black chocolate or entirely of white chocolate.

Luka would rather start eating as soon as possible instead of wasting time cutting so he asks you to help him determine the minimum number of cuts required to separate the black squares from the white squares.



Slika 1: This is how the chocolate from the first example should be cut in order to separate the black parts from the white parts with minimum number of cuts

Input

In the first line, there are two natural numbers n and m ($1 \leq n, m \leq 200$), representing the number of rows and columns of the chocolate.

Each of the following n lines contains m characters, each of which is either 0 or 1. The character 0 denotes a white square, and 1 denotes a black square.

Output

In the first and only line, output a single number - the minimum number of cuts required to separate the black squares from the white squares.

Scoring

Subtask	Points	Constraints
1	5	The chocolate is a chessboard, i.e. the square in the i -th row and j -th column is white if $i + j$ is an even number, and black if it is odd.
2	11	$n = 1$
3	11	Exactly one square is black.
4	23	No additional constraints.



Examples

input

4 7
0000000
0111000
0111100
0000000

output

6

input

4 5
00000
01100
01100
00000

output

4

input

4 4
0101
1010
0101
1010

output

6

Clarification of the first example:

Explained in the drawing.

Clarification of the second example:

Luka should make a cut between the first and second row, third and fourth row, first and second column, and third and fourth column.

Clarification of the third example:

Luka should make a cut between every two adjacent rows and columns - a total of 6 cuts.



Task Džeparac

Mother Antonija has earned N euros and must spend all of it as soon as possible. She may keep a portion of the money for herself, but the remaining amount must be equally distributed to her two sons over several days.

First, she chooses a non-negative integer k ($0 \leq k \leq N$) to keep for herself. The remaining amount of $N - k$ euros is then distributed to the sons over d days. Mother Antonija may also choose not to distribute anything to her sons, which corresponds to the case when $N = k$ and $d = 0$.



If the money is distributed, it is done so that each day both sons receive the same amount of money. If one son receives x euros on a given day, the other son also receives x euros, where x must be a positive integer. In total, each son must receive the same total amount of money.

Two distributions are considered different if at least one of the following holds:

- the chosen amount k is different
- the number of days d is different
- there exists at least one day on which the amounts received by the sons differ (i.e., the sequence of daily payments is not identical).

Your task is to determine the number of different ways in which the mother can distribute the money. Since the number of ways can be very large, output the result modulo $10^9 + 7$.

Input

The first line contains a natural number n ($1 \leq n \leq 10^{18}$), the number from the problem statement.

Output

Print a single number - the number of ways in which mother Antonija can distribute the money.

Scoring

Subtask	Points	Constraints
1	12	$n \leq 10$
2	17	$n \leq 1000$
3	36	$n \leq 10^6$
4	5	No additional constraints.

Examples

input

4

output

4

input

5

output

4

input

793

output

137435472

Clarification of the first example: We consider all possible values of the number k :



$k = 4 \rightarrow$ the mother keeps all the money, so the only distribution is the one where the sons receive no money.

$k = 2 \rightarrow$ 2 euros remain to be distributed to the sons. The only valid distribution is $d = 1$, where each son receives 1 euro.

$k = 0 \rightarrow$ 4 euros remain to be distributed to the sons. There are two valid distributions:

$d = 1 \rightarrow$ each son receives 2 euros

$d = 2 \rightarrow$ each day, each son receives 1 euro

$k = 1$ and $k = 3 \rightarrow$ there is no way to divide the remaining amount so that both sons receive the same positive integer amount each day.

In total, there are $1 + 1 + 2 = 4$ different ways of distribution.



Task Prepisivanje

On the eve of an important Croatian language exam, the classroom is already prepared. We can view it as a matrix of dimension $n \times m$, where each cell represents one seat. Each cell of the matrix can have one of three values:

- 2 denotes a seat already occupied by well-behaved students. They are responsible and the teacher is not concerned about them.
- 1 denotes a seat that the teacher has marked as forbidden. No one will sit in those seats.
- 0 denotes an empty seat where mischievous students can sit.



The mischievous students have not arrived yet, but they will come soon. The teacher can decide which empty seats the mischievous students will occupy.

Well-behaved students never cheat, but they can cause mischievous students to cheat if they are adjacent to them. A mischievous student will cheat if they have at least one neighboring student (in one of the four directions: up, down, left, right), whether well-behaved or mischievous.

Determine the maximum total number of students that can be seated in the classroom (including both well-behaved and mischievous students) such that no cheating occurs during the exam.

Input

The first line contains natural numbers n, m ($1 \leq n, m \leq 80$), as described in the problem statement.

Each of the next n lines contains m characters, each being 0, 1, or 2, representing the matrix described in the problem.

Output

In the first and only line, output a single number - the maximum total number of students that can be seated in the classroom such that no cheating occurs during the exam.

Scoring

Subtask	Points	Constraints
1	8	$n, m \leq 4$
2	15	All cells in the matrix will be 0.
3	16	$n = 2$
4	52	$n \leq 15$
5	19	No additional constraints.



Examples

input

4 4
0100
0202
1000
2120

output

6

input

4 4
0000
0000
0000
0000

output

8

Clarification of the second example: The teacher will seat students in the first and third rows in odd-numbered columns, and in the second and fourth rows in even-numbered columns. This way, she will ensure that no student can cheat.



Task Skijanje

Skier Mia spent a day at an unusual ski resort. The resort consists of n apres-ski locations (hereafter: apres) connected by slopes such that they form a connected tree rooted at the apres labeled 1. Each slope is directed from the apres with a smaller number to the apres with a larger number.

The tree is defined as follows: for each apres $i > 1$, it is known from which apres p_i ($p_i < i$) one arrives at it, i.e., its parent in the tree is known. This uniquely determines all slopes (slope i leads to the apres labeled i).



During the day, Mia skied each slope exactly once and for each slope remembered its fun factor z_i and speed b_i .

At the end of the day, Mia wants to ski down once more. Since Mia is very tired from skiing all day, she will choose a run consisting of at most k consecutive slopes. The run must follow the direction of the slopes (from apres with smaller numbers to those with larger numbers). After finishing the run, a helicopter will pick her up and take her home.

For any chosen run, we define its hecticness as follows. Let:

- z_{first} be the fun factor of the first slope in the run
- z_{last} be the fun factor of the last slope in the run
- b_i be the speeds of all slopes in that run

Then the hecticness of the run is: $z_{\text{last}} \cdot (z_{\text{last}} + \sum b_i) + z_{\text{first}}^2$. In the case $k = 1$, the formula remains unchanged, and $\text{first} = \text{last}$ holds.

Your task is to determine the maximum hecticness of a run that Mia can ski.

Input

The first line contains natural numbers n, k ($1 \leq k \leq n \leq 3 \cdot 10^5$), as described in the problem statement.

The second line contains $n - 1$ integers, where the i -th number represents from which apres one can reach the apres labeled $i + 1$ ($1 \leq p_i \leq i$).

The third line contains $n - 1$ integers, where the i -th number represents the fun factor of the $(i + 1)$ -th slope ($1 \leq z_i \leq 10^5$).

The fourth line contains $n - 1$ integers, where the i -th number represents the speed of the $(i + 1)$ -th slope ($-10^5 \leq b_i \leq 10^5$).

Output

In the first and only line, output a single number - the maximum hecticness of a run that Mia can ski.

Scoring

Subtask	Points	Constraints
1	14	$n \leq 1000$
2	23	For each $1 \leq i < n$ it will hold that $z_i = 1$ and $b_i > 1$.
3	35	$n \leq 50000$
4	38	No additional constraints.



Examples

input

```
5 1
1 2 2 1
5 4 8 7
6 3 9 3
```

output

```
200
```

input

```
9 2
1 2 1 1 4 3 6 5
1 3 7 8 4 1 8 2
1 -7 -1 -6 3 8 -1 6
```

output

```
120
```

Clarification of the first example: Since $k = 1$, it follows that the maximum hecticness of a run is equal to the maximum hecticness among the individual slopes. The slopes have hecticness values of 80, 44, 200, and 119. The maximum hecticness is 200.



Task Učionica

A group of k friends enters a classroom. The height h_i of each friend is known, and the heights of all other students already in the classroom are also known.

The classroom can be represented as an $n \times m$ grid a , where each cell corresponds to a seat. The seat in the i -th row and j -th column is denoted by $a_{i,j}$. For every seat, we know whether it is already occupied by a student (along with their height) or if it is empty.

The k friends want to take seats in the classroom. Each friend occupies exactly one empty seat, and no two friends can sit in the same seat. Additionally, they want to sit next to each other in the same row. More precisely, they choose a row index i such that $1 \leq i \leq n$ and a column index j such that $1 \leq j \leq m - k + 1$, and sit in the seats $a_{i,j}, a_{i,j+1}, \dots, a_{i,j+k-1}$. The friends can sit in these seats in any order; they do not have to sit in the order in which their heights were given.

A friend considers a seat suitable only if all students sitting in front of them (i.e., in rows with a smaller index in the same column) are strictly shorter than them, ensuring they have a clear view. The group considers a set of k consecutive seats in the same row suitable if they can arrange themselves so that each friend has a clear view.

Considering these conditions, how many suitable sets of seats for the group are there in the classroom?

Input

The first line contains three natural numbers n , m , and k ($1 \leq n, m \leq 2000, 1 \leq k \leq m$) - the number of rows and columns of the classroom and the number of friends.

The second line contains k natural numbers $h_1, h_2, \dots, h_k$ - the heights of the friends.

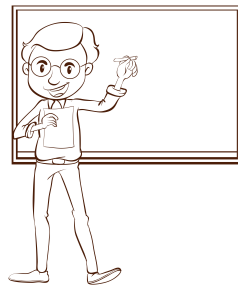
The next n lines each contain m natural numbers: if $a_{i,j} = 0$, the seat is empty, and if $a_{i,j} \geq 1$, the seat is occupied by a student of height $a_{i,j}$ ($1 \leq a_{i,j} \leq 10^9$).

Output

In the first and only line, output a single number - the number of suitable sets of k consecutive seats in the same row where the friends can be arranged so that each has a clear view.

Scoring

Subtask	Points	Constraints
1	11	$k \leq 2$
2	13	$n, m \leq 200$
3	29	$n, m \leq 500$
4	57	No additional constraints.





Examples

input

3 4 2
2 6
0 0 3 1
8 0 0 0
0 0 1 0

output

3

input

2 4 4
5 2 4 3
1 2 3 4
0 0 0 0

output

1

input

5 5 3
17 3 17
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0

output

15

Clarification of the first example:

The two friends can sit in the first and second seats in the first row, in the second and third seats in the second row, or in the third and fourth seats in the second row. So, they can sit in a total of 3 different sets of seats.

Clarification of the second example:

The four friends can sit in the only four available seats if they arrange themselves from shortest to tallest.

Clarification of the third example:

The three friends can sit in any three consecutive seats in the same row since there are no other students in the classroom resulting in a total of 15 valid sitting arrangements.